EFU SHIPPING LICENCE v1.0

Experimental EFU-based Sea, River, Freight, and Passenger Shipping Framework

0. Purpose and Scope

Experimental EFU-based licensing and measurement framework for all shipping (deep sea, coastal, river, freight, passenger).

NOT an IMO certificate, port fee, or EU MRV replacement, but an open research protocol measuring shipping's physical footprint in human-scale EFU metrics (fuel, maintenance, ballast, emissions, port infrastructure). Covers ocean liners, container ships, barges, ferries, and cruises in unified framework.

I. Core Concepts

1. **EFU-Ship (EFU-SHIPPING)**
   EFU framework application for shipping:
   - **EFU-H_SHIP**: bunker fuel/heavy fuel oil/LNG
   - **EFU-M_SHIP**: hull paint, antifouling, propeller wear
   - **EFU-C_SHIP**: CO_2, SOx, NOx, black carbon
   - **EFU-B_SHIP**: ballast water exchange/organism transfer
   - **EFU-INFRA_SHIP**: port dredging, berth amortization
2. **Shipping Sovereignty Gap (SS_SHIPPING)**
   SS_SHIPPING = TEFU_SHIPPING_total – Regional Shipping Carrying Capacity (RSCC_SHIPPING)
   Measures if shipping volume stays within regional port/route regeneration capacity.
3. **MROI_SHIPPING**
   MROI_SHIPPING = THI_SHIPPING / TEFU_SHIPPING
   THI_SHIPPING = global trade connectivity + passenger mobility + cargo throughput efficiency.

II. EFU Shipping Licence – Research Status

Documents EFU-audited shipping lines, port operators, barge fleets, cruise operators. **Does NOT replace** safety certificates, MARPOL compliance, or ballast water management conventions.
Enables Maersk-scale container benchmarking vs. Rhine/Danube barges.

III. Metrics and Thresholds

1. EFU Components (per TEU-km or passenger-km)

EFU-H_SHIP (Fuel)
EFU-H_SHIP = fuel_MWh / 12.88 / TEU_km or pax_km
Example: Container ship = 0.02 liter/TEU-km → 0.00005 EFU/TEU-km

EFU-M_SHIP (Maintenance)
EFU-M_SHIP = antifouling_paint_kg / 30kg/person/year / TEU_km

EFU-C_SHIP (Emissions)
EFU-C_SHIP = CO2e_kg / 4t/person/year / TEU_km

EFU-B_SHIP (Ballast)
EFU-B_SHIP = ballast_m3 / 1000m3/person/year / voyage

TEFU_SHIPPING = 1.2×EFU-H + 1.5×EFU-M + 1.3×EFU-C + 1.0×EFU-B + 0.9×EFU-INFRA

2. Regional Shipping Carrying Capacity (RSCC_SHIPPING)

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RSCC_SHIPPING = Sustainable_cargo_billion_TEUkm / Population / 15ktkm/person/year
Example: EU = 1.2T TEUkm / 448M / 15ktkm = 0.18 EFU_SHIPPING/person/year
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IV. Licence Categories (EFU-SHIPPING Labels)

Level	SS_SHIPPING	MROI_SHIPPING	Label	Interpretation
1	≤ 0	≥1e-2	**EFU-SHIPPING SOVEREIGN**	Within port/route capacity
2	0 < ≤25k EFU	≥5e-3	**EFU-SHIPPING BALANCED**	Monitorable
3	25k < ≤125k EFU	≥1e-3	**EFU-SHIPPING DEFICIT**	Port congestion risk
4	>125k EFU	<1e-3	**EFU-SHIPPING CRITICAL**	Global supply chain strain

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V. Regional RSCC_SHIPPING Calibrations

Region	Cargo (B TEUkm/year)	Population (M)	RSCC_SHIPPING (EFU/person/year)
EU	1,200	448	0.18
USA	800	340	0.16
CHINA	2,500	1,410	0.12

Region	Cargo (B TEUkm/year)	Population (M)	RSCC_SHIPPING (EFU/person/year)
SINGAPORE	200	6	0.48
NETHERLANDS	150	18	0.46

VI. Example: EU Container Shipping (Maersk/Rotterdam)

Input (annual sector):

- TEU-km: 1.2T TEUkm
- Fuel: 120M tonnes HFO (400B MWh)
- Antifouling paint: 50,000 tonnes
- Crew: 200,000 seafarers
- Ports: Rotterdam/Antwerp (Netherlands RSCC_SHIPPING = 0.46 EFU/person/year)

Calculation:

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EFU-H_SHIP = 400B/12.88/1.2T = 0.000026 EFU/TEUkm
EFU-M_SHIP = 50k/30/1.2T = 0.0000014 EFU/TEUkm
EFU-C_SHIP = 500M tCO2e/4/1.2T = 0.00010 EFU/TEUkm

TEFU_SHIPPING_total ≈ 180,000,000 EFU/year
Allocated = 200k × 0.46 = 92,000 EFU/year
SS_SHIPPING = 180M – 92k = +179,908,000 EFU/year (Level 4: CRITICAL)

Result: EFU-SHIPPING CRITICAL
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Sub-sector Comparison: Rhine barges = 0.15 EFU/TEUkm (Level 2: BALANCED)

VII. Implementation Notes

1. **IMO MRV alignment**: TEFU_SHIP as DCS (Data Collection System) enhancement
2. **EU ETS shipping**: EFU-C_SHIP for weighted emissions in cap-and-trade
3. **Port congestion pricing**: EFU-INFRA_SHIP basis for dynamic berth fees
4. **Data sources**: Tier 1 = AIS + bunker delivery notes; Tier 2 = IMO GISIS + port statistics